

August 17, 2026

Dear Andrea Gervasio and Systematix Hiring Team,

I am writing to apply for the Data Scientist contract role with Systematix. With a Master of Science in Computer Science from the University of Calgary and hands-on experience in Python, large-scale data processing, predictive modelling, model evaluation, documentation, and technical communication, I am excited by the opportunity to contribute to your client's optimization and predictive modelling initiative.

My graduate research focused on scalable machine learning and data-management workflows for large geospatial and Earth observation datasets. I designed, trained, and evaluated deep neural networks in Python and PyTorch, built reproducible preprocessing and evaluation pipelines, and documented assumptions, limitations, and results. This work required careful attention to data quality, sampling strategy, computational performance, and the trade-offs involved in turning analytical goals into reliable implementations.

One of my primary research projects examined the reliability and generalizability of deep-learning models for digital elevation model super-resolution. I developed Python-based data workflows, implemented training and evaluation pipelines, and measured performance using statistical and terrain-specific metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors. Through systematic experimentation, I found that random data splitting could produce overly optimistic results and designed region-based validation methods that provided a more realistic assessment of model behavior. That experience is directly relevant to building models that are accurate, understandable, stable, and defensible.

I have also built C++ and Python software for processing, encoding, storing, and retrieving large volumes of spatiotemporal satellite data, improving one encoding workflow from 233 seconds to 26 seconds using C++ multi-threading. These projects strengthened my understanding of data architecture, ETL-style workflows, performance analysis, reproducibility, and the practical challenges of making large datasets usable for downstream modelling.

The role's focus on mixed-integer linear programming and optimization solvers is an area where I would be building further depth, but the surrounding responsibilities match my background strongly: Python, Pandas-style data work, predictive modelling, large datasets, debugging analytical behavior, documenting assumptions, and communicating methods and trade-offs clearly. I have also worked with SQL, PostgreSQL, Docker, Git, Linux, AWS-based coursework, and backend APIs, which gives me a practical engineering foundation for maintainable data science workflows.

I also bring experience explaining complex analytical concepts to different audiences. As a teaching assistant at the University of Calgary and a Deep Learning Teaching Assistant with Neuromatch Academy, I supported students through programming, data, and machine learning workflows, reviewed project work, and helped teams reason through ambiguity. These experiences strengthened my ability to write clear documentation, surface risks early, and support knowledge transfer.

Thank you for considering my application. I look forward to the opportunity to discuss how my Python, modelling, data-processing, documentation, and research background can contribute to this Systematix client engagement.

Sincerely,
Amir Mirzai Golpayegani
amirgolpaa24@gmail.com